UNIVERSAL BOUNDEDNESS OF FIBRES DESCENDS ALONG REPRESENTABLE MORPHISMS WHOSE IMAGE CONTAINS THE SOURCE IMAGE

THE STACKS PROJECT AUTHORS

Editorial note. The source ends '(Some details omitted.)'; the omitted step is the standard transfer of finiteness and the degree bound along the displayed field-extension base change, carried as an explicit obligation.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 380 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Points-cartesian surjectivity. Properties of Spaces, lemma points-cartesian (cross-chapter, named). Lifts a field point of X through f to a field point of Y.
- Definition of universally bounded fibres. definition-universally-bounded (same chapter, structural). Fixes the degree bound n for the base-changed morphism.

Source reference. Original definition 03JL: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 277–277. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H4: focused-seeds/geometry/S9-decent-descent-universally-bounded/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $g : Y \to X$ be a representable morphism of algebraic spaces over S. Let $f : U \to X$ be a morphism from a scheme towards X. Let $f' : U \times_X Y \to Y$ be the base change of f. If*

$$\mathrm{Im}(|f| : |U| \to |X|) \subset \mathrm{Im}(|g| : |Y| \to |X|)$$

and f' has universally bounded fibres, then f has universally bounded fibres.

Proof. Let $n \geq 0$ be an integer bounding the degrees of the fibre products $\mathrm{Spec}(k) \times_Y (U \times_X Y)$ as in Definition 03JL for the morphism f'. We claim that n works for f also. Namely, suppose that $x : \mathrm{Spec}(k) \to X$ is a morphism from the spectrum of a field. Then either $\mathrm{Spec}(k) \times_X U$ is empty (and there is nothing to prove), or x is in

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

the image of $|f|$. By Properties of Spaces, Lemma 03H4 and the assumption of the lemma we see that this means there exists a field extension k'/k and a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(k') & \longrightarrow & Y \\ \downarrow & & \downarrow \\ \mathrm{Spec}(k) & \longrightarrow & X \end{array}$$

Hence we see that

$$\mathrm{Spec}(k') \times_Y (U \times_X Y) = \mathrm{Spec}(k') \times_{\mathrm{Spec}(k)} (\mathrm{Spec}(k) \times_X U)$$

Since the scheme $\mathrm{Spec}(k') \times_Y (U \times_X Y)$ is assumed finite of degree $\leq n$ over k' it follows that also $\mathrm{Spec}(k) \times_X U$ is finite of degree $\leq n$ over k as desired. (Some details omitted.) $\square$